Aditya Jaiswal

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Education

Imperial College London — London, United Kingdom
MSc in Applied Computational Science and Engineering | Grade: Ongoing — 2026–2027

- Modules: Computational Math, Data Science & Machine learning, Deep Learning, Scientific Machine Learning

Birla Institute of Technology and Science, Pilani — Pilani, India
B.E. in Mechanical Engineering | Grade: First Class — 2021–2025

- Modules: Calculus (A+), Complex Analysis & Linear Algebra (A), Probability & Statistics (A-), Differential Equations (A-)

Notable Achievements

JEE Mains:	Top 1% amongst the 10 million candidates who took this exam.	2021
JEE Advanced:	Top 5% amongst the 0.15 million candidates who took this exam.	2021
INSPIRE Scholarship:	Awarded to the top 1% students in India's national 12 grade school exams.	2021

Work Experience

ideaForge — Mumbai, India
R&D — Oct 2025 – May 2026

- Processed UAV test log data in **Python** and automated qualification Excel-sheet updates using **openpyxl** and **xlwings**, speeding up analysis from **30 minutes to 3 seconds** (**600x** speedup); adopted team-wide.
- Led in-house propeller development, trading off **thrust, weight, and efficiency** through simulations, and making the final call on component sourcing.

Indian Institute of Science, IISc — Bengaluru, India
Undergraduate Thesis – Computational AeroSciences Lab — Aug 2024 – Jan 2025

- Designed a priority-based **greedy mesh-recombination algorithm**, parallelizing the search in **Python** with **MPI** to reduce degrees of freedom by **23%** while scaling to large meshes.
- Achieved a **93%** quad-conversion rate through an optimized element-selection strategy implementing carefully chosen quality functions.
- Built a custom mesh-format converter and cross-framework validation pipeline, ensuring reproducibility across independent simulation systems and environments.

Indian Institute of Technology Bombay, IITB — Mumbai, India
Summer Research Fellow — May 2023 – Aug 2023

- Applied **asymptotic perturbation analysis** to derive closed-form correction terms for a numerical PDE solver, implemented in **C++**, correcting systematic bias in a physics simulation.
- Derived higher-order correction factors up to $O(\epsilon^2\alpha^2)$ via analytical modeling; validated against benchmark tests, **reducing model error** from **1.7%** to **<0.1%** and **8%** to **0.5%** across two numerical schemes.

Conference papers, Poster presentation & Publication

– Malyadeep Bhattacharya, **Aditya Jaiswal**, Amol Subhedar, Improved surface tension force scheme for two-phase flow in diffuse interface framework, **COMPFLU-2023**, 18th–20th Dec 2023, IIT Madras.

Projects

Black–Scholes Option Pricing using Lattice Boltzmann Method — Jul 2026 – Aug 2026

- Achieving an **RMSE of** 2.0×10^{-3} against the closed-form analytical price, close to a **Crank–Nicolson** finite-difference benchmark (5.4×10^{-4}).
- Solving the Black–Scholes PDE with a **D1Q3 Lattice Boltzmann** scheme in **Python** by transforming it into a heat-diffusion equation, after diagnosing and correcting a boundary-term bias in the initial formulation.

Technical Skills

Languages & Software C/C++, Python, BASH (shell), Matplotlib, MS Excel